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**RV COLLEGE OF ENGINEERING®**

(An Autonomous Institution affiliated to VTU)

I / II Semester B. E. Regular / Supplementary Examinations Feb – 2024

COMMON TO AI, BT, CS, CY, CD, IS

**QUANTUM PHYSICS FOR ENGINEERS****Time: 03 Hours****Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 & 11 are compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10, and 11 lab components (compulsory).
3. Physical constants:  $h = 6.625 \times 10^{-34} Js$ ,  $m_e = 9.1 \times 10^{-31} kg$ ,  $k_b = 1.38 \times 10^{-23} J/K$
4. Use of Physics formula handbook is permitted.

**PART A**

|   |      |                                                                                                                                                                                                                      |    |
|---|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1 | 1.1  | What is the difference between Boltzmann factor and Boltzmann constant?                                                                                                                                              | 01 |
|   | 1.2  | Why does laser need an active medium?                                                                                                                                                                                | 01 |
|   | 1.3  | Why does the wave function of a particle in an infinite well go to zero at the two boundaries?                                                                                                                       | 01 |
|   | 1.4  | What is the momentum of a particle in the 1st excited state of an infinite well of size $a$ according to de-Broglie's principle?                                                                                     | 01 |
|   | 1.5  | A 2 qubit state is represented by a matrix of what dimension?                                                                                                                                                        | 01 |
|   | 1.6  | Write the matrix representation of a $CNOT$ gate.                                                                                                                                                                    | 01 |
|   | 1.7  | The Fermi factor of an energy level is 0.42. What is the probability that this energy level is higher than the Fermi energy?                                                                                         | 01 |
|   | 1.8  | For $InAs$ at $10^0 C$ , calculate the ratio of the number of states per unit energy per unit volume at two energies which are $36 meV$ and $16 meV$ above the bottom of the conduction band ( $m_e^* = 0.18 m_e$ ). | 01 |
|   | 1.9  | Type I superconductors at $T > T_c$ , belong to which Electric phase?                                                                                                                                                | 01 |
|   | 1.10 | Give one application of AC Josephson Effect.                                                                                                                                                                         | 01 |

**PART B**

|   |   |                                                                                                                                                                                                                                                                       |    |
|---|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 2 | a | Starting with the classical wave equation first derive an expression for Time Independent Schrodinger equation and then apply the relevant boundary conditions to arrive at the normalized wave function for a particle in an infinite well extending from 0 to $a$ . | 10 |
|   | b | A spectral line intensity $I_0$ broadens by $10 pm$ . If its life time is $0.1 \mu s$ , calculate the central wavelength of the broadened spectral line and the wavelength at which the intensity of the spectral line becomes $I_0/2$ .                              | 04 |
| 3 | a | Explain the concept of a Qubit with the help of a neat diagram of the Bloch sphere. What is the action of the Pauli $X$ gate to a point on the Bloch sphere? Prove it by acting the Pauli $X$ matrix on the $ket 0\rangle$ .                                          | 06 |

|   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
|---|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|   | b | Calculate the inner product of two states given as : $ \psi\rangle = 0.9659 \phi_1\rangle + 0.2588 \phi_2\rangle$ and $ \chi\rangle = 0.5 \phi_1\rangle + 0.866 \phi_2\rangle$ . What is the angle between them?                                                                                                                                                                                                                                                            | 08 |
|   |   | <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| 4 | a | With a neat figure explain the principle, construction and result of the single photon interference experiment.                                                                                                                                                                                                                                                                                                                                                             | 10 |
|   | b | Calculate $\sigma_x\sigma_y + \sigma_y\sigma_x$ . This is also known as the anti-commutator.                                                                                                                                                                                                                                                                                                                                                                                | 04 |
| 5 | a | Derive an expression for the concentration of electrons in the conduction band for an intrinsic semiconductor. If the band-gap is halved, how much will the electron concentration change? (Assume $E_f = E_g/2$ ) Given $E_g = 0.4\text{eV}$ , $T = 1159\text{K}$ .                                                                                                                                                                                                        | 10 |
|   | b | A rectangular plate of semiconductor having a width of $2\text{mm}$ is subjected to a current of $4\text{mA}$ along the length and a magnetic field of $2\text{Tesla}$ , oriented parallel to the slab thickness. If the Hall voltage is $2\text{mV}$ , calculate the Hall coefficient of the material and its charge carrier concentration.                                                                                                                                | 04 |
|   |   | <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| 6 | a | Diagrammatically show the variation of the Fermi level as<br>i) A function of varying doping for fixed temperature and<br>ii) A function of varying temperature for fixed doping.<br>Explain the reason for this variation for both the cases.                                                                                                                                                                                                                              | 10 |
|   | b | The Fermi energy for copper is $7.0\text{eV}$ . For copper at $227^\circ\text{C}$ , determine the energy of the electron level for which the probability of occupation is 0.95.                                                                                                                                                                                                                                                                                             | 04 |
| 7 | a | Describe the 3 processes when light interacts with a 2 level system. With the help of Einstein's A and B Coefficients, derive an expression for the energy density of the electromagnetic spectrum. Derive the relation between A, b by using the Planck's blackbody spectrum relation. IF we take an atomic system with 4 non-equidistant energy levels instead of the hypothetical two levels taken above, how many peaks in the emission spectrum will be there and why? | 10 |
|   | b | The acceptance angle of an optical fiber which has a diameter of $20\mu\text{m}$ is $30^\circ$ . What is the V-number of light of wavelength $6280\text{\AA}$ is passed through it? Hence calculate the number of modes for a step index fiber.                                                                                                                                                                                                                             | 04 |
|   |   | <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| 8 | a | The angle with respect to the optical axis, at which light is launched into an optical fiber from air is decreased from $90^\circ$ continuously. Derive an expression for the maximum angle for which the light gets transmitted. If we continue to decrease the angle of incidence, what would happen to the intensity of transmitted light and why?                                                                                                                       | 10 |
|   | b | The incident angle $\theta$ at the air-core interface is taken small such that $\sin\theta = \theta$ (in radians). Calculate the geometrical length traversed by the ray before it has its first total internal reflection in terms of $\theta$ , $d$ and $n_c$ . $d$ =diameter of the optical fiber and $n_c$ is the refractive index of the core respectively.                                                                                                            | 04 |

|                                     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                          |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
|-------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------------|-------------------------------|------|---|---|---|-------------------------------------|-----|-----|-----|------|------|------|-----|--|--|--|---|----|-----|--|--|--|---|----|-----|--|--|--|---|----|-----|--|--|--|----|
| 9                                   | a               | Discuss with diagram the construction and working of a charge and flux qubit.                                                                                                                                                                                                                                                                                                                                                                                                                       | 10                       |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
|                                     | b               | The critical field for a Type 1 superconductor at $T = 22^0K$ is found to be 75% of the critical field at $0K$ . Calculate the critical field for $T = 50^0K$ .                                                                                                                                                                                                                                                                                                                                     | 04                       |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
| <b>OR</b>                           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                          |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
| 10                                  | a               | Explain why flux lines are completely expelled from the bulk of a super conductor. Explain the process by which Cooper pairs are formed, according to $BCS$ theory. Explain the difference between Type I and Type II super conductors with the help of M-H diagrams.                                                                                                                                                                                                                               | 10                       |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
|                                     | b               | The critical magnetic field at $15K$ is $2 \times 10^3 A/m$ in a superconductor ring of radius $1cm$ at $T = 20K$ . If the temperature is halved, find the percentage change in the critical current. Given $T_c = 40K$ .                                                                                                                                                                                                                                                                           | 04                       |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
|                                     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                          |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
| 11                                  | a               | With a neat figure, describe the apparatus required and procedure for experimental determination of the wavelength of laser. If the distance between the grating and screen is $1\ meter$ and the diffraction order is given below. The grating constant is $5.08 \times 10^{-5}m$ . What is the wavelength of the laser generated?                                                                                                                                                                 | 10                       |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
|                                     | b               | Complete the given tabular column and plot it on a graph to extract the band gap of a Thermistor, using relevant formula. Write the relevant experimental procedure.                                                                                                                                                                                                                                                                                                                                |                          |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
|                                     |                 | <table><tr><td><i>Diffraction order</i></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td><i>Distance (<math>2X_n</math>)</i></td><td>3</td><td>5.8</td><td>8.7</td><td>11.5</td><td>14.4</td><td>17.2</td></tr></table>                                                                                                                                                                                                                                                 | <i>Diffraction order</i> | 1               | 2                             | 3    | 4 | 5 | 6 | <i>Distance (<math>2X_n</math>)</i> | 3   | 5.8 | 8.7 | 11.5 | 14.4 | 17.2 |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
| <i>Diffraction order</i>            | 1               | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3                        | 4               | 5                             | 6    |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
| <i>Distance (<math>2X_n</math>)</i> | 3               | 5.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 8.7                      | 11.5            | 14.4                          | 17.2 |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
|                                     |                 | <table><tr><td><i>Serial No</i></td><td><i>Temp (C)</i></td><td><i>R(<math>\Omega</math>)</i></td><td></td><td></td><td></td></tr><tr><td>1</td><td>80</td><td>6.7</td><td></td><td></td><td></td></tr><tr><td>2</td><td>75</td><td>7.1</td><td></td><td></td><td></td></tr><tr><td>3</td><td>70</td><td>7.6</td><td></td><td></td><td></td></tr><tr><td>4</td><td>65</td><td>8.1</td><td></td><td></td><td></td></tr><tr><td>5</td><td>60</td><td>8.5</td><td></td><td></td><td></td></tr></table> | <i>Serial No</i>         | <i>Temp (C)</i> | <i>R(<math>\Omega</math>)</i> |      |   |   | 1 | 80                                  | 6.7 |     |     |      | 2    | 75   | 7.1 |  |  |  | 3 | 70 | 7.6 |  |  |  | 4 | 65 | 8.1 |  |  |  | 5 | 60 | 8.5 |  |  |  | 10 |
| <i>Serial No</i>                    | <i>Temp (C)</i> | <i>R(<math>\Omega</math>)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                          |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
| 1                                   | 80              | 6.7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
| 2                                   | 75              | 7.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
| 3                                   | 70              | 7.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
| 4                                   | 65              | 8.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |
| 5                                   | 60              | 8.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |                 |                               |      |   |   |   |                                     |     |     |     |      |      |      |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |   |    |     |  |  |  |    |